Anmol Sharma

Systems & Backend Engineer

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SUMMARY

Backend engineer who builds systems where correctness matters: a deterministic reconciliation engine for messy financial exports, a sandbox that runs untrusted code in locked-down containers, and a hybrid-search RAG service. Specialized in Python and Rust across PostgreSQL, Redis, and Docker, with a rigorous test-driven engineering approach. Minor in Data Science and AI from IIT Mandi (CGPA 8.44/10).

EXPERIENCE

Independent, *AI/ML & Systems Engineer* 04/2024 – Present | Jaipur, India

- Built four backend systems end to end (CanonData, Nimbus, CodexEngine, rust-agent-core): design, implementation, tests and deployment.
- Test coverage: 200+ Pytest cases for CanonData, 40+ for Nimbus, and a RAGAS evaluation harness over a golden query set for CodexEngine.
- Advised a digital marketing agency (The Bullseye) on Python/SQL data-extraction pipelines and n8n webhook workflows for client reporting.

Teleperformance Global Business, *Technical Support Executive* 07/2023 – 10/2023 | Jaipur, India

- Resolved enterprise software and security issues (Norton / Microsoft) against SLA targets.
- Used SQL queries and log analysis to trace application errors and transaction anomalies before escalating edge cases.

EDUCATION

Minor in Data Science and Artificial Intelligence, *Indian Institute of Technology (IIT), Mandi* 05/2025 – 06/2026

Centre for Continuing Education (CCE) | CGPA: 8.44 / 10 Coursework: Mathematics for Data Science (Linear Algebra, Calculus, Optimization), Machine Learning Fundamentals, Deep Learning (Transformers, Attention, CNNs)

Bachelor of Computer Applications (BCA), *Jaipur National University* 06/2019 – 06/2022

Grade: 77.4% — First Division Coursework: Data Structures (C), DBMS, Operating Systems, Computer Networks, OOP (C++)

PROJECTS

CanonData, *Financial Data Reconciliation & Audit Engine*

Commercial B2B System Python, Polars, FastAPI, PostgreSQL, SQLite, SQLAlchemy, Alembic, Next.js, Pytest

- Reads CSV exports from Shopify, Amazon, Meta, Google and others and rebuilds one consistent view of orders, inventory, ad spend and customers. Every figure traces back to its source file, row and column through SHA-256 file hashing and cell lineage; unusable rows are set aside with reason codes instead of dropped.
- Nine leak detectors (phantom ad burn, discounts below cost, zombie returns, frozen working capital and others) sit behind a common Detector interface. Language-model answers must match engine-computed numbers within 0.5% or are replaced by a deterministic template.
- FastAPI service split into 7 routers, with OTP auth and SQLite/PostgreSQL storage. On a synthetic benchmark with 9,578 injected errors (Bramble & Bolt), net sales, product costs, ad spend and stock value land within 2% of the answer key.

Nimbus , *Sandboxed Task Execution Control Plane*

Python, FastAPI, Docker, Redis, WebSockets, Pytest

- Runs each task in a disposable Docker container with all Linux capabilities dropped, no-new-privileges, and limits of 1 GB memory, 2 CPUs and 256 processes.
- Redis/arq worker queue, WebSocket streaming of task output, and a per-task event log stored in the database. 40+ unit and integration tests.

CodexEngine , *Hybrid-Search RAG Service*

Python, LangGraph, FastAPI, PostgreSQL/pgvector, Next.js, RAGAS

- LangGraph pipeline with intent routing, question condensing, retrieval evaluation, and query-rewrite and web-search fallbacks. State is checkpointed in PostgreSQL and answers stream over Server-Sent Events.
- Retrieval combines pgvector similarity with BM25, reranks with a CrossEncoder, and filters by user and thread metadata. Traced with Langfuse; faithfulness and relevancy are scored with RAGAS.

rust-agent-core, *Terminal Agent Runtime*

Rust, Tokio, fastembed-rs (ONNX), Gemini API

- Single-binary CLI with a multi-turn REPL and tool calling (document search, web search, file read/write, shell commands). Tool use is gated by ReadOnly, Safe and Yolo execution modes.
- Generates embeddings locally with ONNX models, so indexing and search need no network call. Falls back to a second Gemini model when the primary fails.

SKILLS

Languages
Python, Rust, SQL, TypeScript

Infra & Observability
Docker, Linux, Vercel, uv, Git, OpenLLMetry (OpenTelemetry), Langfuse

Data
PostgreSQL, pgvector, SQLite, Polars, SQLAlchemy, Alembic, Supabase

Testing & Evals
Pytest, httpx, RAGAS, DeepEval, golden-set evaluation

AI & Retrieval
LangGraph, hybrid search (pgvector + BM25), FAISS, CrossEncoder reranking, ONNX embeddings, tool calling, LLM provider fallback

Backend
FastAPI, Pydantic, REST APIs, JWT/OTP authentication, WebSockets, Server-Sent Events, Redis/arq, Tokio, Next.js (React)

HONORS

Director's Certificate of Appreciation, *Student Council, Symbiosis School for Liberal Arts (SSLA), Pune*

NTSE Scholar, *National Talent Search Examination*